Analytical Synopsis: Portfolio and Investment Analysis

1. Executive Summary

This project evaluates the risk-return profiles of major individual equities (Intel vs. AMD) and mutual funds (Vanguard Growth vs. Value). By utilizing descriptive statistics and formal statistical inference, this analysis provides data-driven investment insights and demonstrates the critical difference between perceived risk and statistically significant risk.

2. Equity Performance Analysis (Intel vs. AMD)

Using recent return data, we calculated key performance and volatility metrics for Intel (INTC) and Advanced Micro Devices (AMD).

* Return and Volatility: Both equities exhibited very low average returns (~0.02). Intel showed a slightly higher standard deviation (0.085) compared to AMD (0.084).

* Risk-Adjusted Performance: The Coefficient of Variation (CV) for both stocks was exceptionally high (INTC: 4.14, AMD: 5.57), indicating that the risk taken for every unit of return is disproportionately large. Most notably, both stocks yielded negative Sharpe Ratios (-0.65 for INTC and -0.72 for AMD). This indicates that, during the observed period, these investments underperformed relative to a risk-free asset.

3. Statistical Inference: Mutual Fund Risk Assessment

A common assumption among investors is that Growth funds are inherently riskier than Value funds. To test this, we applied formal hypothesis testing to the standard deviations of both funds (using n=36 periods).

* Single Population Variance (Chi-Square Test): We first tested if the Growth fund's risk exceeded a client's strict 20% threshold. The calculated test statistic yielded a p-value of 0.052. At a 5% significance level, we do not reject the null hypothesis. The risk does not significantly exceed 20%.

* Two Population Variances (F-Test): To compare the two funds directly, we utilized an F-test. The sample variance difference initially implies higher risk for the Growth fund. However, the F-test statistic resulted in a p-value of 0.051. Because the p-value is greater than our 0.05 significance level, we fail to reject the null hypothesis.

4. Conclusion

The analysis clearly demonstrates the necessity of statistical testing in finance. A cursory glance at the standard deviations leads to the incorrect conclusion that the Growth mutual fund is riskier. Formal testing proves there is insufficient statistical evidence to support that claim. Consequently, an investor should not avoid the Growth fund strictly based on perceived volatility.